

Risk modeling of communications, navigation, and surveillance complex systems of systems for future aviation

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Abstract

The planned integration of Communications (C), Navigation (N), and Surveillance (S) by the U.S. Federal Aviation Administration (FAA) would convert their currently safely coordinated operations into a single vulnerable interdependent and interconnected CNS complex systems of systems (Complex SoS). We model the ensuing vulnerability of planned Complex SoS by tracing specific new interdependencies and interconnections manifested by their shared/common (i) States (variables), (ii) Decisions, (iii) Decision makers, and (iv) Resources (cyber-physical) among the three C, N, and S systems. Other contributions stem from our exploiting the use of fault trees (without the reliance on reliability data). Exploiting fault trees enabled us to quantify the accrued knowledge of interdependencies and interconnectedness tabulated in Tables 1–4 in terms of AND Gates (systems connected in Parallel), and OR Gates (systems connected in Series); and to discover dangerous shared C, N, and S systems on the critical path. Namely, if all systems (subsystems) in the minimal cut set fail, then the entire CNS Complex SoS would fail. *Significantly, this modeling approach alerted the FAA on “what not to do” by identifying subsystems early in the planned integration of C, N, and S that would result in a single catastrophic failure of the planned CNS Complex SoS.*

KEYWORDS

complex systems of systems, risk modeling, SEE25 systems of systems (SoS)

1 | INTRODUCTION AND OVERVIEW

The Federal Aviation Administration (FAA) is planning the integration of three major systems—Communications (C), Navigation (N), and Surveillance (S) (which today operate semi-independently with coordination) into a single complex systems of systems (CNS Complex SoS). The genesis of the complexity stems from the newly created interdependencies and interconnectedness among the integrated CNS Complex SoS. We trace the interdependencies and interconnections among the integrated CNS in terms of the newly shared/common state (variables) and other “essential entities.” The term “other essential entities” connotes: shared/common decisions, decision makers, stakeholders, resources, time frame, and organizational setups, within and among the three C, N, and S systems. We subsequently develop a sample of four comprehensive tables identifying shared/common States, Decisions, Decision makers, and Resources among all three, members of the CNS Complex SoS. We accomplish this critical/technical task with the help and contributions of FAA and MITRE* domain experts in the operations and interface among members of the CNS Complex SoS. The tables that summarize the interdependencies and interconnections among the three (planned) integration of the C, N, and S systems

provide the essential step in modeling and quantifying the interdependencies and interconnectedness (I-I) resulting from the (planned) systems integration. Subsequent to tabulating the Shared States (Table 1), Shared Decisions (Table 2), Shared Decision makers (Table 3), and Shared Resources (Table 4) within and among the three CNS Complex SoS, we translated the tables into fault trees, without the need for, or reliance on, the reliability of the C, N, and S systems. The tables enable us to determine (with the advice and expertise of FAA operators and technical experts) whether the shared systems, and/or their subsystems are connected in (i) Series (via OR Gates) or in (ii) Parallel (via AND Gates). These essential designations of “AND” or “OR” Gates in the construction of fault trees, which were guided by the expertise and experience of FAA engineers, technicians, and managers, were reconfirmed at a second round. Recall that if all systems (subsystems) in the minimal cut set fail, then the entire CNS Complex SoS would fail. Stated differently, the failure of any subsystem connected in series (as a member of the minimal cut set) would cause the failure of all other connected systems.

In sum, this paper introduces a theoretically based methodology to model and identify safety critical interdependent and interconnected Complex SoS. Limiting our modeling efforts to only those prospective

faults/risks initiated at one subsystem that can propagate to other subsystems of the Complex SoS (due to shared states and other essential entities) captures the most significant risks to the entire planned integrated CNS Complex SoS.

With future technologies and a fully integrated CNS Complex SoS, the current level of assumed safety would significantly change. Clearly, such an unprecedented systems integration of three safety-critical systems into a single CNS Complex SoS must undergo extensive risk modeling, assessment, management, and communication review prior to any partial or full deployment of the integrated system. The following is an overview of the theoretical and methodological processes for modeling CNS Complex SoS that were used by the MITRE/University of Virginia team on this FAA case study.

2 | THEORETICAL AND METHODOLOGICAL CONTRIBUTIONS

This paper develops a quantitative modeling methodology for interdependent and interconnected cyber-physical Complex SoS. We build on two major tenets of systems modeling and risk analysis, namely (i) the central role of state variables in modeling and (ii) supported by exploiting the traditional use of fault-tree analysis, without the reliance on reliability. We achieve this by task, first by identifying the I-I that characterize the interface between and among the planned integrated C, N, and S systems and their associated subsystems. We build on the fact that all outputs from a system are functions of the states of that system, and thus also of the decisions and exogenous variables, and all other inputs to the system. This fact is of particular significance to modeling Complex SoS. Chen¹ offers the following succinct definition of *state variable*: “The state $x(t_0)$ of a system at time t_0 is the information at t_0 that together with the input $u(t)$, for $t \geq t_0$, determines uniquely the output $y(t)$ for all $t \geq t_0$.” The states of a system, commonly a multidimensional vector that characterizes the system as a whole, play a major role in estimating its future behavior for any given input. Thus, the behavior of the states of the system as a function of time enables modelers to determine, under certain conditions, the system's future behavior for any given *input*, or *initiating event* (e.g., to determine the reliability and functionality of a car, one must know the *state* of the fuel, oil, tire pressure, and other mechanical and electrical components). All systems are characterized at any moment by their respective state variables and the conditions thereof, and these conditions are subject to continuous variation and fluctuation. We embrace the fact that complex systems cannot, by their essence and definition, be understood or modeled using one “straight-jacket” modeling schema. Most importantly, by understanding what makes I-I so central to the complexity of SoS, and challenging to both understanding and modeling them, we will ultimately improve future management of Complex SoS.

We discover and model the essence of their complexities by tracing their I-I via shared/common states and other essential entities shared among the individual systems and their subsystems. The time frame of a system is a major driver of what we term complexity, because all systems continue to evolve, emerge, and change, while our ability to keep pace with

these changes, via modeling, continues to lag. Although, our analytical analyses assume a steady-state model, our domain-expert FAA/MITRE teams did project likelihood of future evolving I-I within and among the subsystems of the planned integrated CNS Complex SoS.

A key step in the modeling process was the identification (with several aviation domain experts) of critical shared (i) States, (ii) Decisions, (iii) Decision makers, and (iv) Resources among the three systems and their subsystems (see Tables 1–4, respectively). Clearly, without projecting and understanding, and thus tabulating the consequences resulting from the ensuing shared states and other essential entities, the planned integration of the three semiautonomous systems could result in disastrous consequences. To focus on the most consequential shared states and other essential entities, we deployed risk filtering, ranking, and management (Haimes et al., 1992; Haimes, 2016), the inclusion of which is beyond the scope of this paper. Specific details on fault trees and their essential contributions to the theory and methodology for quantifying the I-I among the three C, N, and S systems via fault trees are introduced in Section 4.2. With our nontraditional use of fault-tree analysis, we are able to quantify the I-I of the shared states and other essential entities that result from the integration of the three C, N, and S systems into a single CNS Complex SoS. We achieve this task by exploiting the use of fault trees without neither the need of knowledge, nor the reliability, of the systems under consideration. Thus, we were able to: (i) Discover the consequential interdependencies and interconnections that would emerge from the planned integration of the three major safety-critical systems into a CNS Complex SoS realized by developing Tables 1–4 of shared/common States, Decisions, Decision makers, and Resources within and among the three C, N, and S systems and their subsystems; (ii) Quantify the risk of dire consequences resulting from the I-I of shared states and other essential entities by connecting shared states and other essential entities with AND Gates (systems connected in parallel) and OR Gates (systems connected in series); and (iii) Exploit the use of fault trees (without a required knowledge of the reliability of each system or subsystem) through modeling using AND Gates, and OR Gates. Clearly, an effective modeling of any system, particularly Complex SoS, requires a clear understanding of its configuration. This methodological approach enabled us to identify and track the essence of the system's complexity via the common/shared states and other essential entities among the systems that compose the CNS Complex SoS.

3 | WHAT HAVE WE LEARNED FROM THE OPEN LITERATURE?

The literature on modeling Complex SoS has been growing rapidly. Most of the early publications on large-scale systems focused on decomposition and multilevel optimization. Examples include: *New Directions in General Theory of Systems*,² *Multilevel Optimization*,³ *General Systems Theory*,⁴ *Optimization Theory for Large Systems*,⁵ *Methodology for Large Scale Systems*,⁶ and *Systems Theory: Philosophical and Methodological Problems*.^{7–9} In addition, there have been some articles on the problem of risk modeling of SoS, many of which have emerged within the last decade. For example, Maier,¹⁰ Sage and Cuppan,¹¹

and Jamshidi,¹² among many others, provided valuable insight into SoS and definitions of emergent behavior of Complex SoS. More recently, Maier¹³ addressed the role of modeling and simulation in SoS development; in which he identified five key lessons learned. Ottino¹⁴ reviewed three major tools for quantitative modeling and studying complex systems: nonlinear dynamics, agent-based models, and network theory. Shalizi¹⁵ reviewed the main methods and techniques of complex systems, which include tools for analyzing data, constructing and evaluating models, and measuring complexity. Chang and Harrington¹⁶ provided a comprehensive description of agent-based models of organizations. Bar-Yam¹⁷ suggested the necessity of considering the specific organizational and technological requirements needed to perform effectively in a high-complexity environment. DeLaurentis,¹⁸ Lewe et al.,¹⁹ Parker,²⁰ and Dahmann and Baldwin²¹ all suggested that SoS problems require a new modeling paradigm that can account for the multiplicity of stakeholders, objectives, interdependencies, and emergent outcomes. Fisher²² argued that emergent behavior is inherent in SoS, and that traditional software and systems engineering methods are inadequate for the interpretation of SoS.

The essential characteristics of SoS present serious difficulties for traditional hazard and risk analysis methodologies. Bristow et al.²³ argued that risk analysis of extreme events affecting SoS should address complex, ambiguous, and uncertain aspects of extreme risk, and the strategic interactions among multiple participants. Perrow²⁴ pointed out that the root causes of system accidents reside in the properties of complex systems themselves, rather than in the errors that owners and operators make in running them—that it is the system's characteristics that make it inherently vulnerable to such accidents. Haimes²⁵ and Guo and Haimes^{26,27} addressed risk modeling of Complex SoS by building on state-space theory. Eusgeld et al.²⁸ discussed the potential failure propagation among infrastructures leading to cascade failures, and they analyzed two modeling alternatives, comparing integrated to coupled models.

Clearly, no review of the literature can do justice to our understanding and modeling of Complex SoS and the associated risks thereto, topics that have occupied the interest of so many scholars and practitioners over the last half-century.

4 | MODELING AND QUANTIFYING THE INTERDEPENDENCIES AND INTERCONNECTEDNESS CHARACTERIZING COMPLEX SYSTEMS OF SYSTEMS

4.1 | Modeling shared states and essential entities in complex systems of systems: Specific lessons learned

The risk modeling efforts are centered on identifying shared states and other essential entities between at least two of the three C, N, and S systems. Tables 1–4 provide safety-critical knowledge about the added sources of risk resulting from the I-I among the components of the CNS Complex SoS. According to Tables 1–4, (i) five states are shared between communications and navigation, (ii) three states are shared between communications and surveillance,

TABLE 1 Shared states (system view)—dynamic RNP configuration

States	Communication	Navigation	Surveillance
HF signal integrity	✓	✓	
VHF signal integrity	✓	✓	
UHF signal integrity		✓	✓
SATCOM signal integrity	✓	✓	
GPS signal integrity		✓	✓
GPS signal availability		✓	✓
GPS signal accuracy		✓	✓
Radar signal integrity			✓
Comprehension	✓		
Plane authenticity/identification	✓		✓
Signal effective range	✓	✓	✓
Station authenticity/identification		✓	
Distance calculation accuracy		✓	
Position drift		✓	
Detection availability			✓
Compliance	✓	✓	✓

TABLE 2 Shared decisions—dynamic RNP configuration—approach

Decision	Communication	Navigation	Surveillance
Type of approach			
Instrument flight rules	✓	✓	✓
Visual flight rules	✓	✓	
Approach system			
ILS		✓	
Vectored approach	✓		✓
GPS (RNAV) approach		✓	
VOR approach		✓	
NDB approach		✓	
Runway clearance	✓		
Aircraft separation	✓		✓
Approach route	✓	✓	
Holding pattern	✓	✓	

(iii) six states are shared between navigation and surveillance, and (iv) one state (compliance) is shared with all three systems. Using Tables 1–4, we are able to determine the most critical shared states and other essential entities. Together, they further our understanding of each of the C, N, and S systems, as well as their interdependencies.

For example, the increased utilization of Global Positioning System (GPS) technology has enabled the performance of advanced activities, such as dynamic required navigation performance (Dynamic RNP), resulting in a significant gain in efficiency and savings. At the same time, there is an added source of risk associated with using a Dynamic RNP configuration when relying on GPS technology. This is because GPS and Dynamic RNP efficiencies have significantly increased the interdependencies within and among the CNS Complex SoS components. More specifically, Table 1 represents shared safety-critical states and their

TABLE 3 Shared decision makers—dynamic RNP configuration—approach phase

Decision\decision maker	Pilot	ATC	Airport
Type of approach			
Instrument flight rules	✓	✓	
Visual flight rules	✓	✓	
Approach system			
ILS	✓	✓	✓
Vectored approach	✓	✓	✓
GPS (RNAV) approach	✓	✓	✓
VOR approach	✓	✓	✓
NDB approach	✓	✓	✓
Runway clearance		✓	✓
Aircraft separation		✓	
Approach route	✓	✓	
Holding pattern		✓	

TABLE 4 Shared resources—dynamic RNP configuration—approach phase

Resource	Communication	Navigation	Surveillance
Radar sites			✓
DME substations		✓	
Satellites	✓	✓	
Air traffic controllers	✓		✓
Pilots	✓	✓	
Airplane	✓	✓	✓
NACO charts	✓	✓	✓
Flight management computer	✓	✓	
ATN – B2 receiver/transceiver	✓		
NAVAID receiver		✓	
Weather stations	✓	✓	✓

place within the integrated CNS Complex SoS. *In a subsequent discussion, we will refer to Tables 1–4 as critical precursors to catastrophic failures due to the (i) Shared states (Table 1), (ii) Shared decisions (Table 2), (iii) Shared decision makers (Table 3), and (iv) Shared resources (Table 4).*

The use of GPS in both the navigation and surveillance systems make GPS a vital resource, if not an essential entity. Indeed, the integrity, availability, and accuracy of the GPS become vital to the integrated CNS Complex SoS. Furthermore, via the shared states and their essential entities, we are able to discover the criticality and integrity of the very-high frequency (VHF) signal, which connects the communication and navigation systems through the signals used in both voice and data communication and navigation. The effective signal range is vital to all three systems and indicates the importance of complete coverage and an understanding of when and where to switch from one system to another. Furthermore, navigation is largely considered the pilot's responsibility and is aided by the flight management system (FMS). Surveillance, considered the air traffic controller's (ATC) responsibility, is the ability to determine the aircraft's position

within the airspace and maintain correct separation among all aircraft. The communication system represents the connection between the navigation and surveillance systems and can be separated into two categories—voice communication and data communication—and modeled accordingly. The current FAA system relies heavily on voice communication between the pilot and the ATC. While this link will never fully disappear with the CNS integration, the new system will rely much more heavily on data communication. Voice communication is used to vector aircraft, avoid weather, maintain specific separation, ask for and receive altitude corrections, and other pilot-controller communication. In the current data communication system, the aircraft communications addressing and reporting system is used to push small text data to and from the aircraft. The communication is limited to initial flight plan loading, text messages, weather updates, and terminal conditions.

Furthermore, the integration of three safety-critical systems into one CNS Complex SoS, also required systematic, comprehensive answers to the following basic questions in risk analysis *for each of the three C, N, and S systems, and for the entire integrated CNS Complex SoS*: (i) What can go wrong? (ii) What is the likelihood? (iii) What are the consequences? and (iv) What is the time frame? Although our research project did not address the above basic explicit questions in risk analysis (they were delegated to other teams), our results identified critically important sources of risk (via the fault-tree analysis) resulting from the planned integration of the CNS systems into a single Complex SoS.

Assessing and managing the impact of current decisions on future options is necessary in risk modeling and analysis in general, and in particular for safety-critical systems. Each of the three C, N, and S systems currently operates semiautonomously, with the integration period planned to exceed two decades. Thus, the risk-based ramifications of this integration being carried out by a federal agency that has the mandate to *operate and regulate* the entire aviation Complex SoS, cannot be underestimated. Thus, a sample of one of the most consequential findings resulting from our modeling the I-I resulting from the systems integration is the need to systematically examine the shared states and other essential entities with an eye toward “*What not to do.*” *That is, the consequences resulting from the shared states and other essential entities of the C, N, and S systems must be carefully studied and their integration evaluated to avoid “one-point failure” of the aviation Complex SoS.* Our study identified specific aspects of the communications, navigation, and surveillance systems and their subsystems that should never be integrated because their merging could lead to disastrous consequences. (Tables 1–4 represent a sample of our findings). Our theoretical and methodological premises regarding modeling Complex SoS published earlier^{25,29} have been measurably clarified, solidified, and substantiated by this and other case studies. Indeed, this modeling process provided an opportunity for the MITRE/FAA/University of Virginia team to better understand the intricacies resulting from the integration of the C, N, and S systems. This investigative and learn-as-you-go process has lead to (i) developing a better understanding of the consequences resulting from the planned systems integration, (ii) identifying specific safety-critical failure modes associated with the planned integration, and (iii) alerting the decision makers and

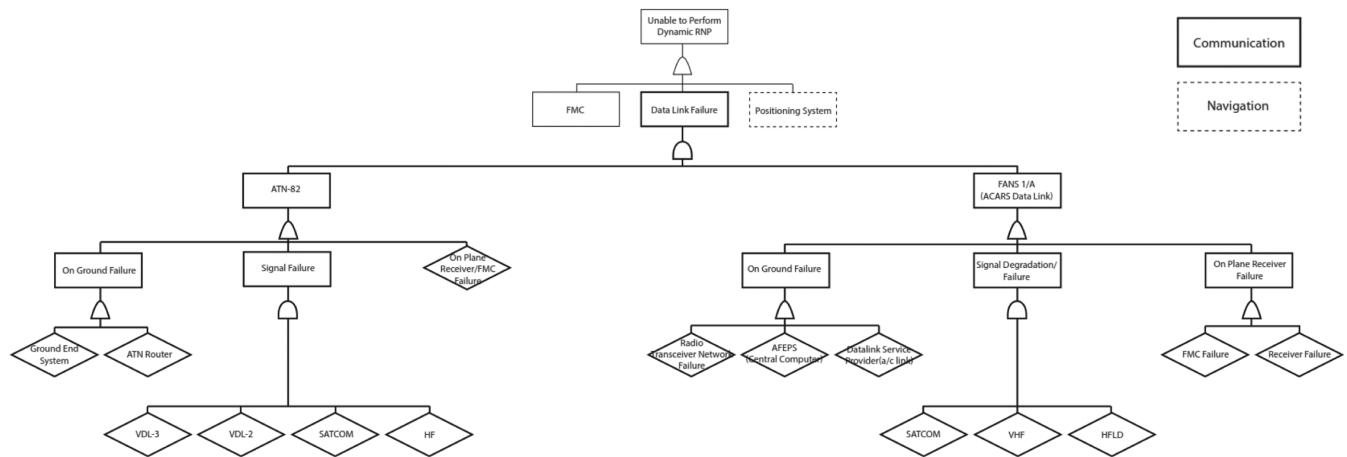


FIGURE 1 Data link fault tree: expanded fault tree with failure modes caused by data link failure

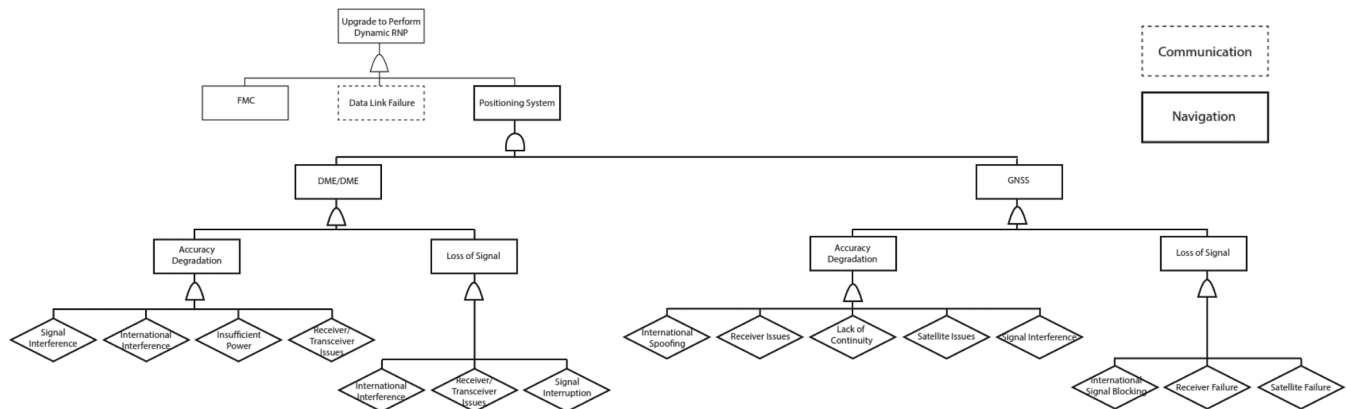


FIGURE 2 Positioning system fault tree: expanded fault tree with failure modes caused by data link failure

stakeholders regarding “what not to do,” probably one of the most important and immediate benefits of the FAA case study.

4.2 | Exploiting the use of fault trees for modeling complexity via shared states and other essential entities

The extensive reliance on fault trees and their exploitation in this paper call for the following succinct description of the fault-tree model,³⁰ which is most relevant to modeling Complex SoS:

A fault tree analysis can be simply described as an analytical technique, whereby an undesired state of the system is specified (usually a state that is critical from a safety standpoint), and the system is then analyzed in the context of its environment and operation to find all credible ways in which the undesired event can occur. The fault tree itself is a graphic model of the various parallel and sequential combinations of faults that will result in the occurrence of the predefined undesired event. The faults can be events that are associated with component hardware failures, human errors, or any other pertinent events, which can lead to the undesired event. A fault tree thus depicts the logical interrelationships

of basic events that lead to the undesired event—which is the top event of the fault tree. It is important to understand that a fault tree is not a model of all possible system failures or all possible causes for system failure. A fault tree is tailored to its top event, which corresponds to some particular system failure mode, and the fault tree thus includes only those faults that contribute to this top event. Moreover, these faults are not exhaustive—they cover only the most credible faults as assessed by the analyst. (p. IV-1)

Failures in engineered systems are primarily triggered by the failure of one or more components (subsystems). Three of the most safety-critical systems of the National Air Space (NAS) are addressed in this case study—Communications, Navigation, and Surveillance. The formulation of policies and procedures (driven by and represented in the “common/shared essential entities”) guide the NAS, the ATCs, the pilots, among myriad other “systems.” Therefore, the complete compliance with them by all parties is essential. Indeed, ignoring the projected consequences resulting from the planned integration of CNS can trigger failures that could lead to unacceptable and catastrophic consequences. Thus, it is imperative that knowledgeable engineers, operators and other experts in the NAS identify all conceivable new sources

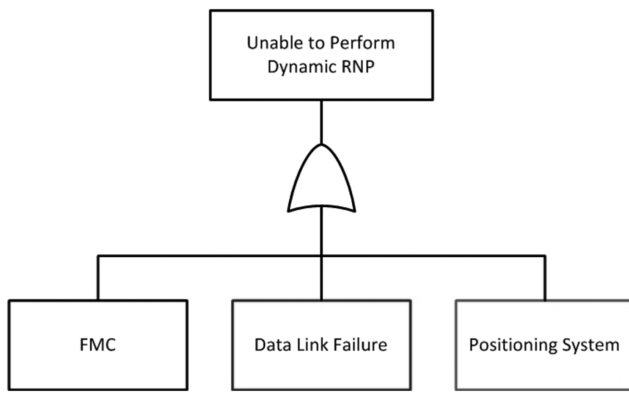


FIGURE 3 Simple dynamic RNP fault tree

of risk (due to the planned integration) that could cause the failure of the safety-critical NAS system.

To better communicate with the readership, we begin by introducing two fault trees, which relate the I-I between communications and navigation systems. Figure 1 is an expanded fault tree with failure modes caused by data link failure in the Data Link fault tree.

Figure 2, an expanded fault tree with failure modes caused by data link failure in the Positioning System fault tree, also highlights the interdependencies between communications and navigation systems, albeit with a focus on the positioning systems.

All fault trees represent the major sources of risks resulting from the I-I among the three C, N, and S systems via the minimum cut sets.

For example, in Figure 1 “On Ground Failure,” one minimum cut set includes the following three systems: (i) radio receiver network failure, (ii) central computer failure, and (iii) data link service provider failure. Similarly, in Figure 2, the minimum cut set “Accuracy Degradation” includes the following five systems: (i) intentional signal blocking, (ii) receiver issues, (iii) lack of continuity, (iv) satellite issues, and (v) signal interference.

In this case study, we address only a sample of shared/common states and other essential entities, each of which could trigger a catastrophic failure. The shared states and other essential entities provide an answer only to the first question in risk analysis: “What can go wrong?” within any of the three C, N, and S systems and with the entire CNS Complex SoS. The four fault trees (presented in Figures 3–6), built on Tables 1–4, respectively, illustrate specific relationships of commonalities and I-I among the subsystems of the CNS Complex SoS. The two more comprehensive fault trees (Figures 1 and 2) will provide the readers of this paper with a more complete view of the encompassing complexity associated with the planned integration of the C, N, and S systems.

The development of the fault tree example (Figure 3) confirms the role and importance of the shared states presented in Tables 1–4 in modeling Complex SoS. For example in Figure 3, the I-I between the following two systems: (i) Communication (i.e., Data Link Failure) and (ii) Navigation (Positioning System) systems indicate their equal requirement in completing the task of flying Dynamic RNP. Thus, when two systems are interconnected in series (as is the case), they will display shared states within the fault tree. The shared states do indeed present

themselves in multiple failure nodes within the fault-tree analysis. *The fact that the Communication and Navigation systems share states; implies that they are connected in series (via OR Gate).* As a specific example, the integrity of the two signals in Table 1 (HF and VHF signals) are members of the critical path in both the Communication and Navigation sections of the fault tree (Figures 5 and 6) and correspond to the shared states previously discovered. Therefore, the fault-tree analysis is harmonious with our initial finding, and “quantifies” the interconnectedness via shared states.

Recall that we structured fault trees to determine and track failure paths associated with shared states, decisions, decision makers, and resources among the integrated CNS Complex SoS. Indeed, fault-tree analysis enabled us to focus on the extreme consequences associated with the planned integration, and on the emergence of new sources of risk of catastrophic failure. We began with the large number of subsystems within the C, N, and S systems, each with its own unique functionality and performance, the reliability of which was neither publicly available nor required for this study. Then, we built on the most valuable attribute of fault tree analysis, namely, identifying and focusing on the newly formed integrated CNS systems identifying the I-I as connected in series (via OR Gates), or in parallel (via AND Gates). Clearly, newly connected CNS systems in series would introduce significant risks to the entire aviation enterprise. As noted earlier, this important discovery was enabled by the FAA domain experts’ identification of safety-critical shared states and other essential entities within the integrated CNS system.

Another important advantage resulting from translating shared states and other essential entities into fault trees is the identification of potential weaknesses caused by the newly formed I-I resulting from the integration of the C, N, and S systems as a Complex SoS. Critical paths to failure were discovered using *minimal cut sets*. By definition, a minimal cut set is a combination of intersections of primary events in parallel sufficient for the top event to occur (if all parallel components fail). This combination is the “smallest” combination in that all the failures in the minimal cut set need to occur for the top event (system failure) to occur. If any one component in the parallel combination does not occur, then the top event will not occur (by this combination).

Each fault tree provides valuable insight into the I-I that could lead system failure. For example, the fault tree in Figure 3 is fully expanded in Figure 4; it shows that the three main systems, the flight management computer (FMC), the data link, and the positioning system, are connected in series, which is necessary for the operation of Dynamic RNP, the FMC, data link, and the positioning system. The Dynamic RNP capability enhances present and future trajectory-based operations by expanding RNP operations to live uplink of RNP values on a leg-by-leg basis, radius to fix legs, and fixed radius transitions. The key is the ability to adjust the route or RNP value in real time, versus a fixed predetermined value and clearance. Each of the three systems, the FMC, the data link system, and the positioning system, when integrated into the CNS Complex SoS, contributes to the minimal cut set of the simple fault tree presented in Figure 3. To determine a minimal cut set for each system, we constructed individual fault trees: the FMC (Figure 4), data link (Figure 5), and positioning (Figure 6). This analysis provides a more comprehensive understanding of the structural

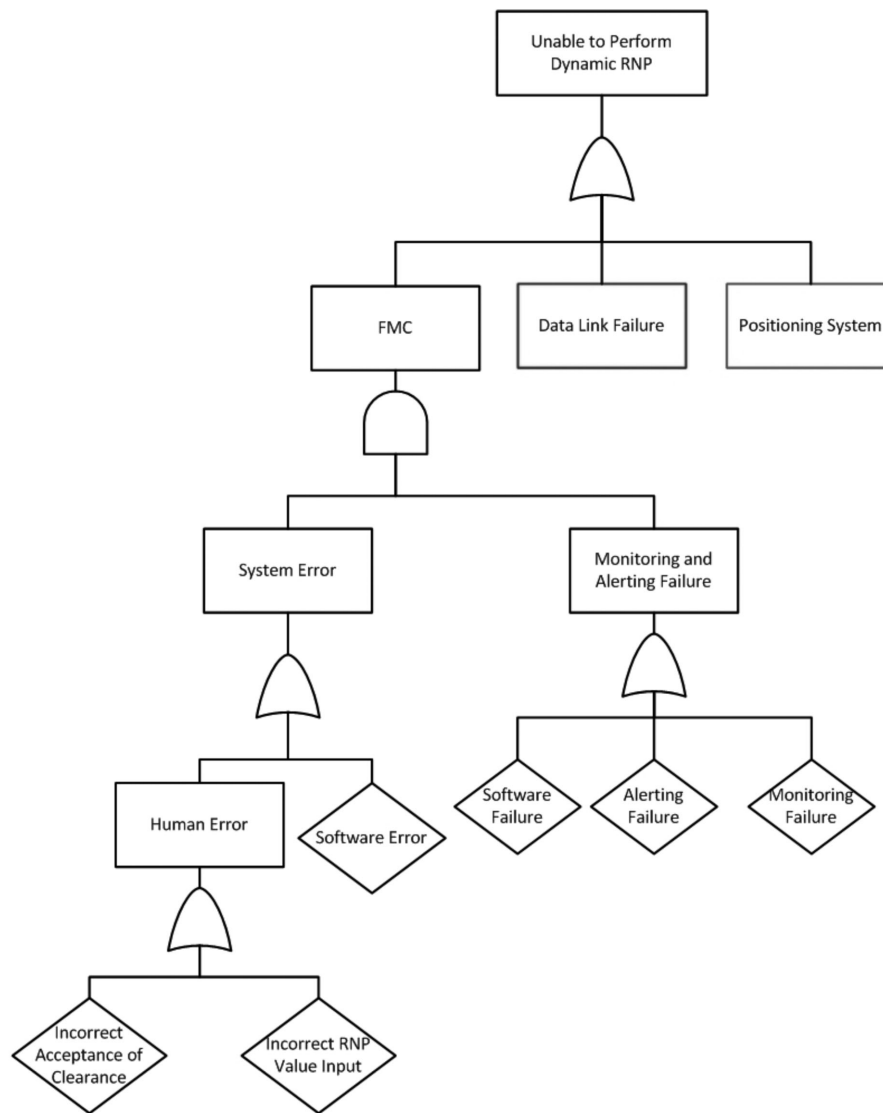


FIGURE 4 The FMC fault tree

perspective of the Dynamic RNP system, and therefore, of the CNS Complex SoS. Furthermore, the fault tree for the FMC (Figure 4) reveals that two subsystems (“system error and monitoring” and “alerting failure”), which are vital to the health of the FMC are connected in parallel via AND Gates.

One of the main benefits of Dynamic RNP is its critical function as a monitoring system to alert and inform the pilot of problems within the system and with the possible inability to maintain the required RNP value for a specific airspace. The multiple OR Gates represented in Figure 4 imply that the minimal cut sets are composed of many combinations of sub-events. For example, one minimal cut set is “incorrect acceptance of clearance and software failure,” another would be “incorrect RNP value input and an alerting failure.” The fault tree for the data link system (Figure 5) is similar to that of the FMC with two main data links available to perform Dynamic RNP. The positioning system fault tree (Figure 6) is similar to the other two main subsystems for the use of the Dynamic RNP capability, where, the GPS global navigation satellite system is considered the main position-

ing system for Dynamic RNP. The fault tree for the FMC (Figure 4) shows two important subsystems vital to the health of the FMC. As these events, “system error and monitoring” and “alerting failure,” are connected with an AND Gate, it will take failures of both events to cause the top event to occur. The OR Gates represent both the “systems error and monitoring” and “alerting failure” events and mean that the minimal cut sets are made up of many combinations of sub-events underneath “system error and monitoring” and “alerting failure” events. Thus, the significance of the minimal cut set and the synergy between this structural perspective and the operational perspective is evident.

In sum, using fault-tree analysis to examine shared states and other essential entities enabled us to determine minimal cut sets and highlight critical I-I within the CNS Complex SoS that would play a key role in the failure of safety-critical systems. Finally, fault-tree analysis and minimal cut sets advance our understanding of, and add insight into, the myriad sources of risk associated with interdependent and interconnected Complex SoS.

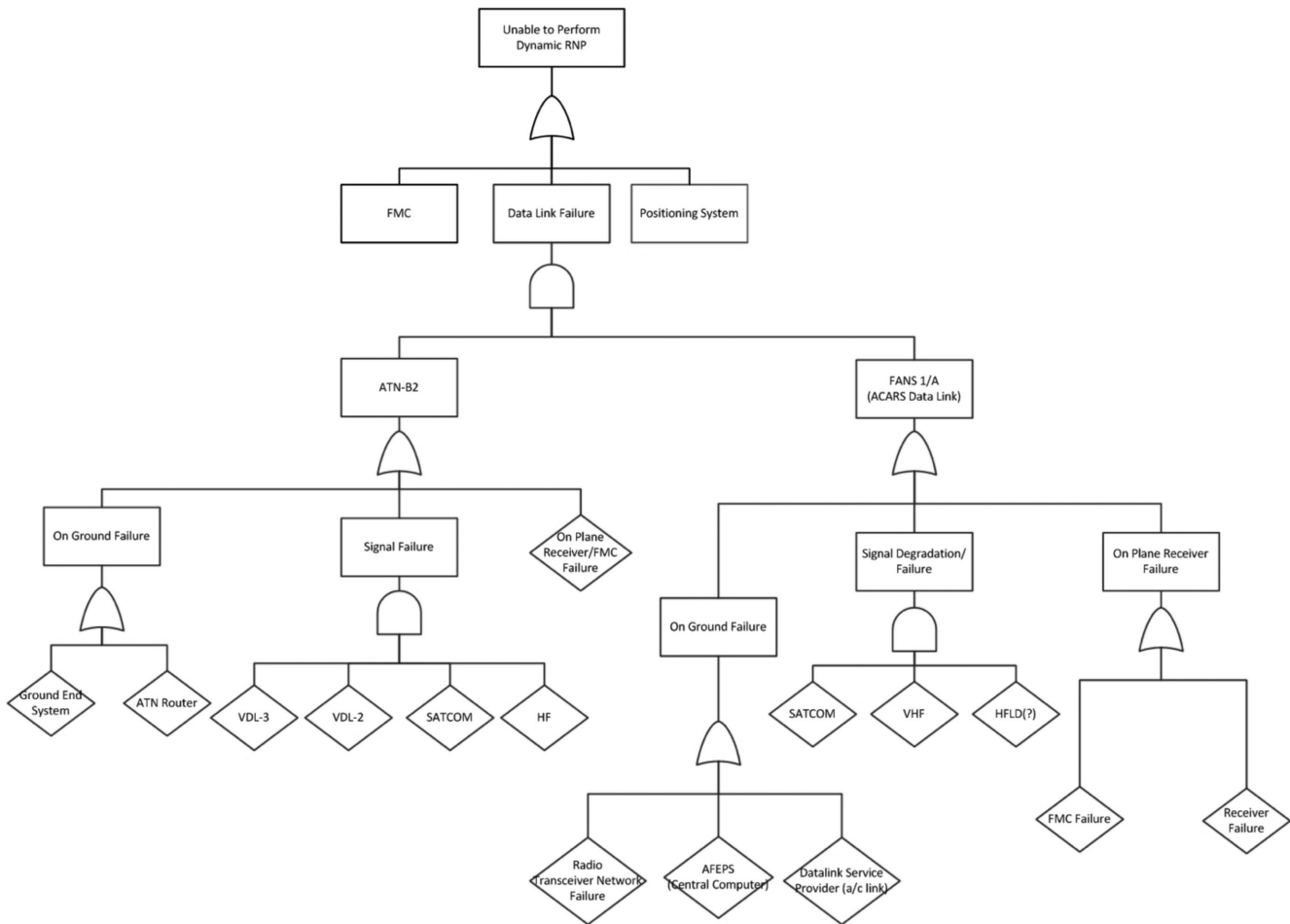


FIGURE 5 Data link failure fault tree

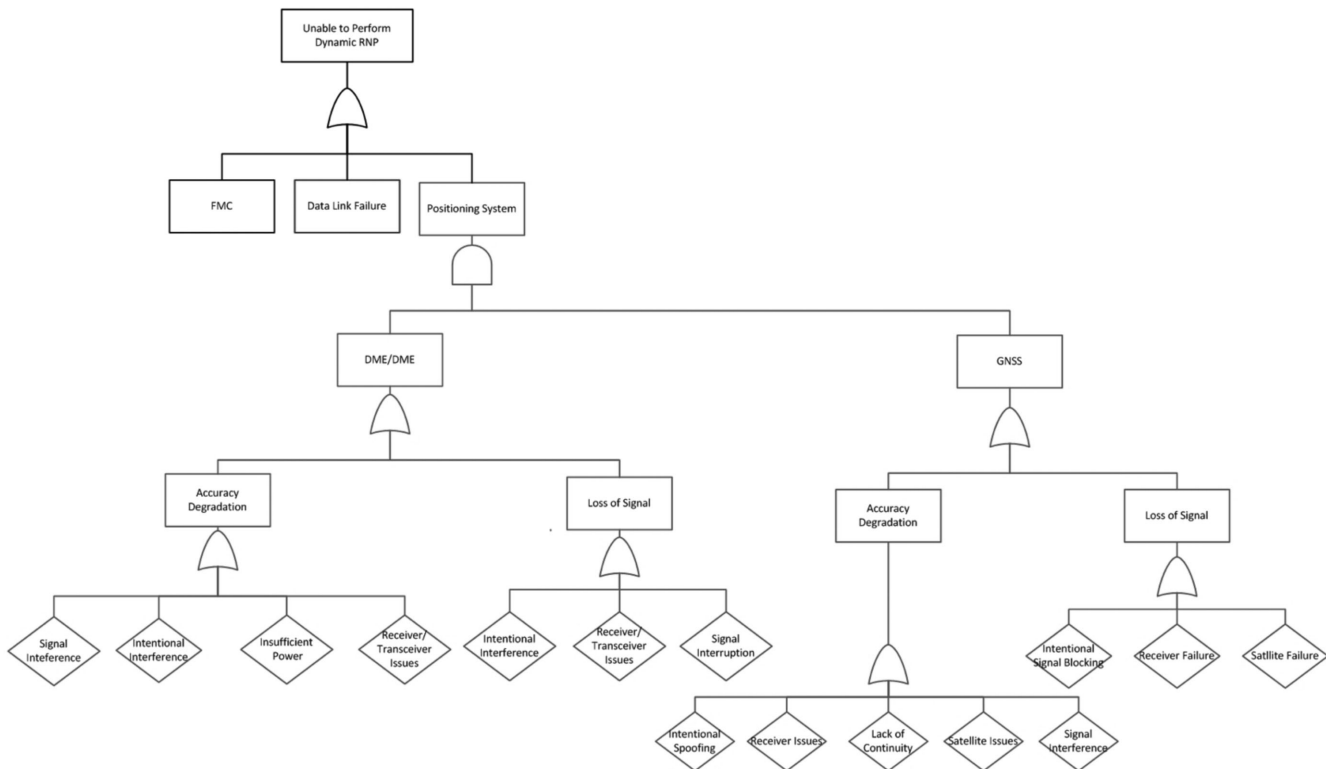


FIGURE 6 Positioning system fault tree

5 | RISK MODELING OF THE CNS CYBER-PHYSICAL SYSTEMS OF SYSTEMS

If the adage “To manage risk, one must measure it” constitutes the compass for risk assessment and management, then modeling constitutes the roadmap that guides the analyst throughout the entire journey of risk analysis. The process of risk assessment and management may be viewed through many lenses depending on one’s perspectives, vision, and circumstances. Kaplan and Garrick (1981)³¹ argued that for each risk scenario in risk assessment we must ask: “What can go wrong? What is the likelihood? What are the consequences?” In the risk management process we ask: “What can be done and what options are available? What are the tradeoffs in terms of all relevant costs, benefits, and risks? What are the impacts of current decisions on future options?” and “What is the time frame?”³² In this case study, not all questions in the risk assessment and management were relevant to the FAA. This position is reasonable because the focus was only on the first question; namely “What can go wrong as a result from merging the three separate systems into a single CNS Complex SoS?” Thus, although all other questions are important, they were too early to be addressed and they were not an integral part of the study. However, if the FAA were interested in pursuing the risk modeling, assessment, management, and communications, then we should have modified the above set of questions from addressing a single system to an interdependent and interconnected Complex SoS: in risk assessment: “What can go wrong for each and every C, N, and S system?” and “What can go wrong for the entire CNS Complex SoS?” The same adjustment should be applied to all other questions. For example, answers to the last question of risk management “What are the tradeoffs in terms of all costs, and benefits?” would require immense effort, given the multiple competing and conflicting objectives associated with the FAA’s Next Generation, of which the integration of CNS is only one part.

6 | SUMMARY AND CONCLUSIONS

We identify two innovations in this paper: (i) Modeling the I-I characterizing Complex SoS by building on shared/common states, decisions, decision makers, and resources within and among the three Communications, Navigation, and Surveillance systems. We subsequently identify and develop comprehensive tables of (i) Shared States, Table 1; (ii) Shared Decisions, Table 2; (iii) Shared Decision makers, Table 3; and (iv) Shared Resources, Table 4. We accomplish this critical/technical task with the support and contributions of FAA domain experts in the operations and interface among members of the planned CNS Complex SoS. The tables that summarize the I-I among the three (planned) integration of the C, N, and S systems provide the essential step in modeling and quantifying the I-I resulting from the (planned) integration the three systems. Our second innovation in the modeling is exploiting the use of fault trees, without the need for, nor reliance on, the reliability on the C, N, and S systems. *The required reliability information for each subsystem in the fault tree is replaced with the shared states, decisions, decision makers, and resources representing the I-I that characterize the planned*

CNS Complex SoS. We relied on the expertise and domain knowledge of the FAA and MITRE teams to populate the fault trees (In the appropriate AND Gates and OR Gates) from the tables (which were also generated with the support of FAA domain-expert engineers). We quantify the I-I by determining whether the systems are connected in (i) Series (via OR Gates), or connected, or in (ii) Parallel (via AND Gates). As noted, this essential designation of “AND” or “OR” Gates in the construction of fault trees is guided by the expertise and experience of FAA engineers, technicians and managers (as well as by the MITRE’s team). Recall that the failure of any subsystem connected in series would cause the failure of all other connected systems; whereas with systems connected in parallel, all subsystems must fail for the entire with system to fail. This fundamental quantitative modeling of the I-I by the exploitation of fault trees enabled the translation of safety-critical I-I into AND Gates and OR Gates and prevent avoidable disastrous consequences that would result from risky integration of subsystems of the CNS systems. In sum, *knowing what not to do via modeling Complex SoS, especially for safety-critical Complex SoS, constitutes an invaluable contribution of systems engineering to public safety.*

ENDNOTE

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